Conservation of Momentum in 2D

Answer key

Conventions used throughout

• **Symbols (from the lessons):** $\vec{p} = m\vec{v}$ is momentum (kg·m/s). A prime (′) means *after* the collision; no prime means before. Subscripts name the object.

• **Method (from the lesson):** draw a before and after diagram, choose the + direction, then write $\vec{p}_{tot} = \vec{p}_{tot}'$, that is $m_1v_1 + m_2v_2 = m_1v_1' + m_2v_2'$.

• **Signs:** in one dimension every velocity carries its sign in the working ($v = -3.0$ m/s). The direction is written in square brackets only in the final answer.

• **Two dimensions (from the lesson):** momentum is conserved in each direction separately. Split every momentum into x- and y-components, conserve each, then combine the components with the Pythagorean theorem and tan θ.

• **Directions:** east is +x and north is +y. Final directions are written like [S 52° E]: start at south and turn 52° toward east.

• **Solve in letters first:** each unknown is isolated as a formula before any numbers go in.

• $g = 9.81$ m/s^2. Every step keeps full precision; only final answers are rounded.

2D COLLISIONS

Four collisions between billiard balls

The diagrams below show several possible two-dimensional collisions that might occur between two billiard balls.

Draw arrows on the diagrams to show how the billiard balls would move after the collision.

Briefly explain your reasoning below:

Totally Inelastic: $m_A = m_B$, $v_A = v_B$

Totally Inelastic: $m_A = m_B$, $v_A \gg v_B$

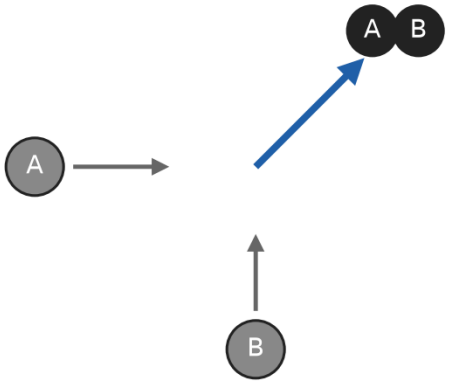
Totally Inelastic: $m_A \gg m_B$, $v_A = v_B$

Elastic: $m_A = m_B$, $v_A = v_B$

In every diagram A moves to the right and B moves up the page, so A brings only x-momentum (m_Av_A) and B brings only y-momentum (m_Bv_B). The total momentum after the collision must have those same two components.

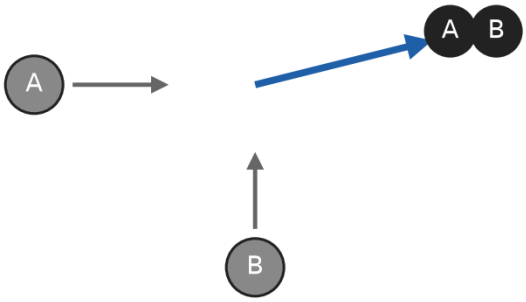
After each collision (grey = before, blue = after)

Totally inelastic $m_A = m_B,\ v_A = v_B$



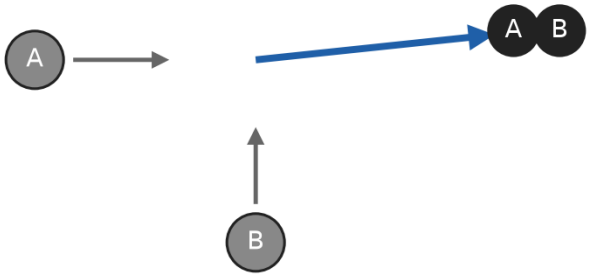
45° up from A's direction

Totally inelastic $m_A = m_B,\ v_A \gg v_B$



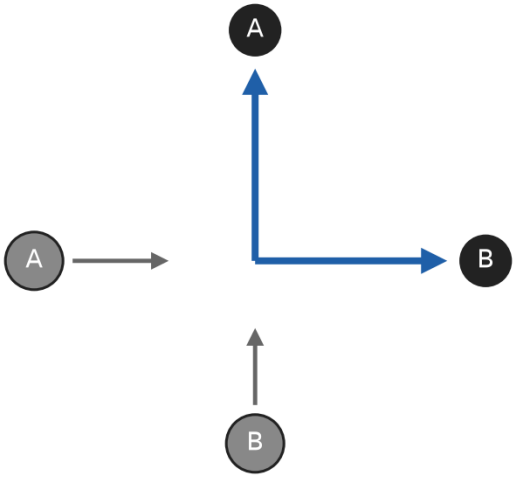
mostly along A's direction

Totally inelastic $m_A \gg m_B,\ v_A = v_B$



almost exactly A's direction and speed

Elastic $m_A = m_B,\ v_A = v_B$



the balls separate: A up, B right

Collision	After the collision	Reasoning
Totally inelastic, $m_A = m_B$, $v_A = v_B$	Stuck together, moving at 45° between the two original directions, slower than before (about 0.71v)	The x- and y-momenta are equal, so the total points at 45°. The pair has twice the mass, so each velocity component is v/2.
Totally inelastic, $m_A = m_B$, $v_A \gg v_B$	Stuck together, moving mostly in A's direction, tilted slightly toward B's direction, at about half of A's speed	A's momentum is much larger than B's, so the total points almost along A's direction. The mass has doubled, so the speed is about half.
Totally inelastic, $m_A \gg m_B$, $v_A = v_B$	Stuck together, moving almost exactly as A was moving	A's momentum is again much larger. B adds very little mass or momentum, so A's velocity barely changes.
Elastic, $m_A = m_B$, $v_A = v_B$	The balls bounce apart. A leaves moving up the page and B leaves moving to the right, each with speed v	The total momentum (45° between the two directions) and the total kinetic energy must both stay the same. Two equal balls that meet like this trade velocities.

FINAL ANSWER

Inelastic, equal masses and speeds: together at 45°

Inelastic, $v_A \gg v_B$: together, mostly along A's direction

Inelastic, $m_A \gg m_B$: together, almost unchanged from A's motion

Elastic, equal masses and speeds: the balls separate and trade velocities

INQUIRY

PhET Collision Lab in two dimensions

Numbers not given: the momenta in 1 and 2 come from your own run of the simulation, so the key cannot give them. It gives what they must show.

1. Before clicking on “play”, do the following:

   a) Click on the “More Data” button at the left side. Record the total x-momentum and the total y-momentum

   b) Look at the Momentum Diagram and note the location of the pink Total Momentum arrow.

2. Click play to run the collision. After the collision, allow the balls to move a few cm apart on the screen. Click on pause. Record the final x and y-momenta. Note the location of the pink arrow

3. Compare your answers for a) and b). What do you notice?

The total x-momentum after the collision equals the total x-momentum before it, and the same is true for the total y-momentum. The pink Total Momentum arrow has the same length and direction before and after. The individual balls’ arrows change, but they always add to the same total.

FINAL ANSWER

3. x_{tot} and y_{tot} are each the same before and after: momentum is conserved in both directions

EXAMPLE

Grenade in three pieces

A 1.2 kg grenade is at rest on a frictionless surface when it explodes into three pieces. A 0.5 kg piece flies off along the surface at 3.0 m/s [N] and a 0.3 kg piece flies off at 4.0 m/s [SW]. What is the velocity of the third piece?

a) What is the total initial momentum of the system? _____

b) On a separate page, sketch a diagram showing the momentum of the three pieces. Show the x- and y-components of each

c) Use conservation of momentum to determine the x- and y-components of the third piece's momentum. Show your work.

d) Combine the components to find the total final momentum and then find the velocity.

GIVEN

$m_{tot} = 1.2$ kg	$m_1 = 0.5$ kg	$\vec{v}_1' = 3.0$ m/s [N]
$m_2 = 0.3$ kg	$\vec{v}_2' = 4.0$ m/s [SW]	$\vec{v}_3' = ?$

Sign convention: east is +x and north is +y. [SW] is 45° between south and west, so both of its components are negative.

a) What is the total initial momentum of the system?

Zero. The grenade is at rest before it explodes, so $\vec{p}_{tot}$ = (1.2 kg)(0) = 0. The three momenta after the explosion must also add to zero, in x and in y.

b) sketch a diagram showing the momentum of the three pieces. Show the x- and y-components of each

Find the size of each known momentum first. Piece 1 points north, so it has only a y-component. Piece 2 points southwest, so it has equal negative x- and y-components.

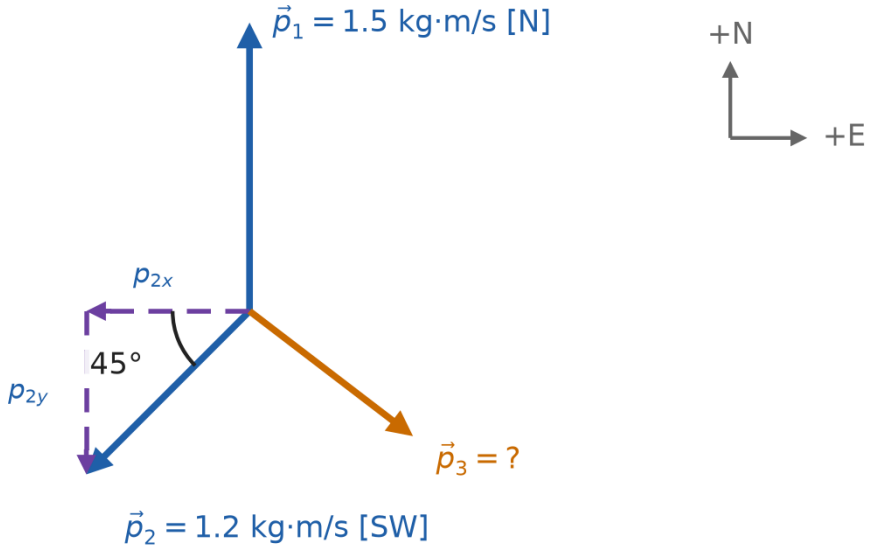
Piece 1: $p_1 = m_1 v_1' = (0.5)(3.0) = 1.5$ kg·m/s [N]

Piece 2: $p_2 = m_2 v_2' = (0.3)(4.0) = 1.2$ kg·m/s [SW]

Piece 2, x: $p_{2x} = -(1.2)\cos 45° = -0.8485$ kg·m/s

Piece 2, y: $p_{2y} = -(1.2)\sin 45° = -0.8485$ kg·m/s

Momentum of the three pieces (east is +x, north is +y)



Piece 3 must point south and east so that the three momenta add to zero.

c) Use conservation of momentum to determine the x- and y-components of the third piece's momentum.

The total is zero in each direction.

x-momentum: $0 = p_{1x} + p_{2x} + p_{3x}$

Isolate p_{3x}:

$$p_{3x} = -p_{1x} - p_{2x} = -(0) - (-0.8485) = +0.8485 \text{ kg·m/s}$$

y-momentum: $0 = p_{1y} + p_{2y} + p_{3y}$

Isolate p_{3y}:

$$p_{3y} = -p_{1y} - p_{2y} = -(1.5) - (-0.8485) = -0.6515 \text{ kg·m/s}$$

p_{3x} is positive (east) and p_{3y} is negative (south).

d) Combine the components to find the total final momentum and then find the velocity.

Step 1 – Size of the momentum of piece 3

Pythagorean theorem:

$$p_3 = \sqrt{p_{3x}^2 + p_{3y}^2} = \sqrt{(0.8485)^2 + (0.6515)^2} = 1.0698 \text{ kg·m/s}$$

Step 2 – Mass and speed of piece 3

The three pieces add up to the 1.2 kg grenade.

Mass: $m_3 = 1.2 - 0.5 - 0.3 = 0.4$ kg

Speed: $v_3' = \dfrac{p_3}{m_3} = \dfrac{1.0698}{0.4} = 2.6744$ m/s

Step 3 – Direction

The angle is measured from south toward east, so the east component is the opposite side.

Formula: $\tan\theta = \dfrac{p_{3x}}{|p_{3y}|}$

Put in the numbers: $\theta = \tan^{-1}\left(\dfrac{0.8485}{0.6515}\right) = 52.48°$

FINAL ANSWER

a) $\vec{p}_{tot} = 0$

c) $p_{3x} = +0.85$ kg·m/s, $p_{3y} = -0.65$ kg·m/s

d) $\vec{v}_3' = 2.7$ m/s [S 52° E]

Worksheet answer: $\vec{v}_3' = 2.67$ m/s [S 52° E]
Why the answers differ: Same value, different rounding. The key gets 2.6744 m/s. The speeds (3.0 and 4.0 m/s) have 2 significant figures and the masses (0.5 and 0.3 kg) only 1, so the key keeps 2 figures, 2.7 m/s; the worksheet keeps 3.